EVERY INVERTIBLE MODULE ON A LOCALLY NOETHERIAN SCHEME WITHOUT EMBEDDED POINTS HAS A REGULAR MEROMORPHIC SECTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-regular-meromorphic-section-exists-noetherian is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 6348-6353 and proof span 6355-6363 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof writes the section as a product over the generic points; the author's notation for these local generators is preserved.

Editorial note. The hypothesis 'having no embedded points' is exactly what makes the product description applicable, and the author states it in the lemma as written.

Editorial note. 1 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The description of the sheaf of meromorphic functions and of meromorphic sections of an invertible module as products of skyscrapers at the generic points, established by a previous card of this lane. Cited at proof lines 6358-6360. Turns a family of local generators at the generic points into a global meromorphic section.
- At a generic point of a locally Noetherian scheme the local ring is artinian, so a rank one free module over it is generated by any nonzero element. Used at proof lines 6355-6358. Produces the local generators used to build the section.

Source reference. Original lemma 0EMH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18307–18307. Same-chapter reference used in the retained author text at source line(s) 6361. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let X be a locally Noetherian scheme having no embedded points. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Then $\mathcal{L}$ has a regular meromorphic section.*

Proof. For each generic point η of X pick a generator s_η of the free rank 1 module $\mathcal{L}_\eta$ over the artinian local ring $\mathcal{O}_{X,\eta}$. It follows immediately from the description of $\mathcal{K}_X$ and $\mathcal{K}_X(\mathcal{L})$ in Lemma 0EMH that $s = \prod s_\eta$ is a regular meromorphic section of $\mathcal{L}$. □